Statistical Modeling and Risk Factor Analysis of Cardiovascular Disease Using Multivariable Logistic Regression

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***Abstract*—Cardiovascular disease (CVD) is a leading cause of mortality worldwide, and early identification of at-risk individuals is essential for preventive intervention. This paper presents a statistical and machine-learning-based analysis of ten-year coronary heart disease (CHD) risk using the Framingham Heart Study dataset (4,240 records, 16 attributes). Missing values in glucose, total cholesterol, BMI, heart rate, cigarettes per day, education, and blood-pressure medication were imputed using median and mode strategies. Formal hypothesis testing — independent-samples t-tests, chi-square tests of independence, Pearson correlation, and one-way ANOVA — was used to identify statistically significant risk factors, followed by a multivariable logistic regression (via statsmodels) to estimate adjusted odds ratios with 95% confidence intervals. Age, systolic blood pressure, cigarettes per day, glucose, male sex, and prior stroke emerged as statistically significant independent predictors of CHD ($p < 0.05$). A separate scikit-learn logistic regression classifier trained on an 80/20 stratified split achieved 84.8% test accuracy, though its recall on the minority CHD-positive class remained low (5%) due to substantial class imbalance (84.8% negative vs. 15.2% positive), a limitation discussed alongside recommendations for resampling and alternative evaluation metrics.**

***Index Terms*—cardiovascular disease, coronary heart disease, logistic regression, hypothesis testing, odds ratio, risk factor analysis, Framingham dataset, machine learning**

I. Introduction

Cardiovascular disease (CVD) remains one of the foremost causes of death globally, and coronary heart disease (CHD) in particular accounts for a substantial share of this burden. Identifying the clinical and demographic factors that drive long-term CHD risk allows clinicians to prioritize preventive care and allows data-driven screening tools to be built. The Framingham Heart Study is one of the most widely used longitudinal datasets for this purpose, recording demographic, behavioral, and clinical measurements together with a ten-year CHD outcome label.

This project has two complementary aims. First, it applies formal statistical hypothesis testing to determine which individual risk factors are significantly associated with ten-year CHD risk, and quantifies their adjusted effect using multivariable logistic regression. Second, it trains a supervised classification model to predict CHD risk from patient attributes and evaluates its practical predictive performance. Both analyses are performed on the same cleaned dataset so that the statistical-inference results and the predictive-modeling results can be interpreted together.

II. Dataset Description

The dataset (`heart.csv`, based on the Framingham Heart Study) contains 4,240 patient records and 16 attributes: sex (`male`), age, education level, current smoking status, cigarettes per day, blood-pressure medication use, prior stroke, prevalent hypertension, diabetes, total cholesterol, systolic and diastolic blood pressure, BMI, resting heart rate, glucose, and the binary target `TenYearCHD`. Table I summarizes the missing-value counts prior to imputation.

TABLE I: Missing Values by Feature

Feature	Missing	% of N
glucose	388	9.15%
education	105	2.48%
BPMeds	53	1.25%
totChol	50	1.18%
cigsPerDay	29	0.68%
BMI	19	0.45%
heartRate	1	0.02%

Missing continuous values (`glucose`, `totChol`, `BMI`, `heartRate`, `cigsPerDay`) were imputed with the column median, and missing categorical values (`education`, `BPMeds`) were imputed with the column mode, using scikit-learn's `SimpleImputer`. After imputation, no attribute contained missing values.

The target variable is strongly imbalanced: 84.81% of records are negative (no CHD within ten years) and 15.19% are positive, as shown in Fig. 1. This imbalance is an important factor in interpreting the classification results in Section IV.

III. Methodology

A. Exploratory Data Analysis

Descriptive statistics and a Pearson correlation heatmap (Fig. 2) were computed across all features to inspect distributions and pairwise linear relationships before formal testing.

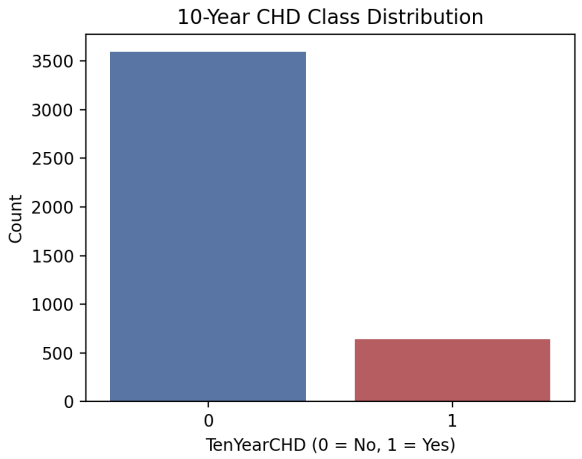


Fig. 1: Ten-year CHD class distribution (84.8% negative, 15.2% positive).

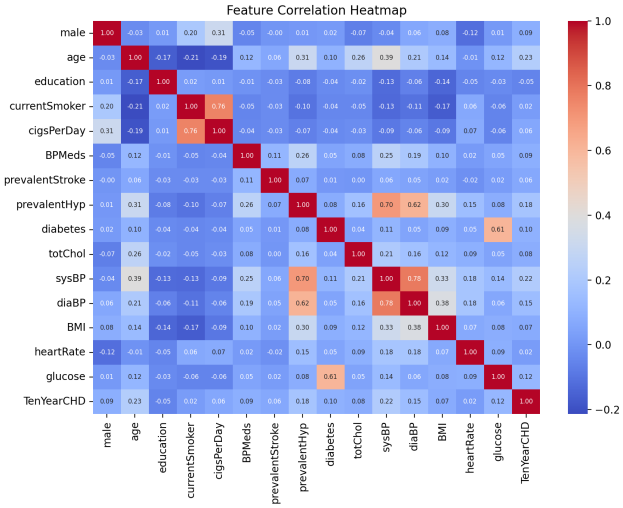


Fig. 2: Pearson correlation heatmap of all features.

B. Hypothesis Testing

Four complementary hypothesis tests were used to screen individual risk factors:

1) Independent-samples t-tests (Welch's, unequal variance) compared the mean of each continuous variable between the CHD-positive and CHD-negative groups (H_0: equal means).
2) Chi-square tests of independence assessed association between each categorical variable and CHD status (H_0: independence).
3) Pearson correlation coefficients quantified the linear association between each continuous variable and the binary CHD outcome (H_0: $r = 0$).
4) A one-way ANOVA tested whether mean total cholesterol differs across the four education levels (H_0: equal means across groups).

A significance threshold of $\alpha = 0.05$ was used throughout.

C. Multivariable Logistic Regression (Inference)

A multivariable logistic regression was fit using `statsmodels` with `TenYearCHD` as the dependent variable and all 15 remaining attributes as predictors (plus an intercept), estimated by maximum likelihood. This model was used purely for statistical inference — to obtain adjusted coefficients, odds ratios, 95% confidence intervals, and p-values for each predictor while holding the others constant — and is distinct from the predictive classifier described next.

D. Predictive Classification Model

A separate scikit-learn `LogisticRegression` classifier was trained for predictive purposes. The data was split into training (80%) and test (20%) sets using stratified sampling on the target variable (`random_state = 42`) to preserve class balance across the split. Model performance was evaluated using accuracy, precision, recall, F1-score, and a confusion matrix on the held-out test set.

IV. Results

A. T-Test Results

Table II reports Welch's t-test results comparing each continuous variable between CHD-negative and CHD-positive patients.

TABLE II: Independent T-Test Results (CHD=0 vs. CHD=1)

Variable	Mean(0)	Mean(1)	t	p
age	48.76	54.15	-15.59	<0.001
totChol	235.13	245.23	-5.01	<0.001
sysBP	130.34	143.62	-12.01	<0.001
diaBP	82.17	86.98	-8.24	<0.001
BMI	25.67	26.51	-4.46	<0.001
heartRate	75.76	76.53	-1.47	0.142
glucose	80.43	88.15	-4.86	<0.001
cigsPerDay	8.65	10.60	-3.55	<0.001

All continuous variables except resting heart rate ($p = 0.142$) show a statistically significant mean difference between the two CHD groups, with CHD-positive patients exhibiting consistently higher values.

B. Chi-Square Test Results

TABLE III: Chi-Square Test Results

Variable	χ^2	df	p
male	32.62	1	<0.001
education	31.05	3	<0.001
currentSmoker	1.50	1	0.221
BPMeds	30.27	1	<0.001
prevalentStroke	14.03	1	<0.001
prevalentHyp	132.46	1	<0.001
diabetes	38.48	1	<0.001

CHD status is significantly associated with sex, education level, blood-pressure medication use, prior stroke, prevalent hypertension, and diabetes (all $p < 0.001$). Current smoking

status alone was not significantly associated with CHD status ($p = 0.221$) in this univariate test.

C. Pearson Correlation Results

TABLE IV: Pearson Correlation with TenYearCHD

Variable	r	p
age	0.225	<0.001
sysBP	0.216	<0.001
diaBP	0.145	<0.001
glucose	0.121	<0.001
totChol	0.082	<0.001
BMI	0.074	<0.001
cigsPerDay	0.059	<0.001
heartRate	0.023	0.137

Age shows the strongest linear association with ten-year CHD risk ($r = 0.225$), followed by systolic blood pressure. All correlations are weak in absolute magnitude but statistically significant, except heart rate.

D. One-Way ANOVA

The one-way ANOVA testing mean total cholesterol across the four education levels was not statistically significant ($F = 1.317$, $p = 0.267$), so the null hypothesis of equal mean cholesterol across education groups is not rejected.

E. Multivariable Logistic Regression (Odds Ratios)

Table V reports adjusted odds ratios (OR) with 95% confidence intervals from the multivariable logistic regression, sorted by statistical significance. The overall model was significant (likelihood-ratio $p < 0.001$; McFadden's pseudo-$R^2 = 0.112$).

TABLE V: Adjusted Odds Ratios (95% CI)

Predictor	OR	95% CI	p
male	1.654	1.358–2.013	<0.001
age	1.064	1.051–1.077	<0.001
sysBP	1.014	1.007–1.021	<0.001
cigsPerDay	1.022	1.010–1.033	<0.001
glucose	1.007	1.003–1.011	0.002
prevalentStroke	2.617	1.102–6.220	0.029
prevalentHyp	1.259	0.979–1.620	0.073
totChol	1.002	1.000–1.004	0.074
BPMeds	1.276	0.829–1.965	0.269
diabetes	1.207	0.678–2.148	0.523

After adjusting for all other variables simultaneously, male sex, age, systolic blood pressure, cigarettes per day, glucose, and prior stroke remain statistically significant independent predictors of ten-year CHD risk ($p < 0.05$). Notably, total cholesterol and prevalent hypertension — significant in the univariate tests — fall short of significance once other correlated predictors are controlled for, and current smoking status, diastolic BP, BMI, heart rate, and education are not significant in the adjusted model.

F. Predictive Model Performance

The scikit-learn logistic regression classifier achieved 84.79% accuracy on the 848-record held-out test set. Table VI shows the full classification report.

TABLE VI: Classification Report (Test Set)

Class	Prec.	Recall	F1	Support
0 (No CHD)	0.85	0.99	0.92	719
1 (CHD)	0.50	0.05	0.10	129
Accuracy			0.85	848
Macro avg	0.68	0.52	0.51	848
Weighted avg	0.80	0.85	0.79	848

The confusion matrix on the test set was $\begin{bmatrix} 712 & 7 \\ 122 & 7 \end{bmatrix}$: of 129 true CHD-positive cases, only 7 were correctly identified (recall $= 5.4\%$), while 712 of 719 negatives were correctly identified. This confirms that although overall accuracy is high, the model performs poorly at its primary clinical purpose — detecting patients who will develop CHD — because it is biased toward the majority (negative) class by the 84.8%/15.2% class imbalance.

V. Discussion

The statistical-inference results and the predictive-model results tell complementary stories. The multivariable logistic regression identifies age, male sex, systolic blood pressure, smoking intensity (cigarettes per day), glucose, and prior stroke as robust, independent risk factors for ten-year CHD, consistent with established cardiovascular epidemiology. These findings are useful for explanatory purposes — understanding which factors drive risk and by how much (via odds ratios) — even though the model's raw classification accuracy is not the primary goal of that analysis.

In contrast, the predictive classifier, while achieving 84.8% accuracy, is not clinically useful in its current form: its extremely low recall (5%) for the CHD-positive class means it fails to flag the vast majority of at-risk patients, defaulting instead to predicting the majority class. This is a direct consequence of the 84.8/15.2 class imbalance and the use of a default 0.5 decision threshold with no class weighting. High accuracy alone is therefore a misleading metric for this task.

Several improvements are recommended for future work: (1) applying class-balancing techniques such as SMOTE oversampling or `class_weight='balanced'` in the classifier; (2) lowering or tuning the decision threshold to trade precision for recall; (3) evaluating with threshold-independent metrics such as ROC-AUC and precision-recall AUC rather than accuracy alone; (4) feature scaling, since the sklearn solver failed to converge within the default iteration limit, indicating unscaled features slow optimization; and (5) exploring regularized or tree-based models (e.g., random forest, gradient boosting) that may better capture non-linear risk interactions.

VI. Conclusion

This study combined formal hypothesis testing with multivariable logistic regression and a supervised classifier to analyze ten-year CHD risk in the Framingham dataset. Age, male sex, systolic blood pressure, cigarettes per day, glucose, and prior stroke were identified as statistically significant, independent risk factors after adjustment. While the trained classifier reached 84.8% overall accuracy, its poor recall on the minority CHD-positive class highlights that class imbalance must be explicitly addressed before such a model could support real clinical screening decisions. Future work should prioritize class-balancing strategies and threshold tuning alongside accuracy to build a model that is both statistically sound and practically useful.

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